Laser Tag 8-Player Hardware Kit

Budget Target: ■1,000 INR (excluding ESP32 microcontrollers)

To outfit 8 players with a gun and a 6-sensor vest while keeping the cost extremely low (~■125 per player), we need to source components in bulk and use clever, low-cost materials for the physical gear.

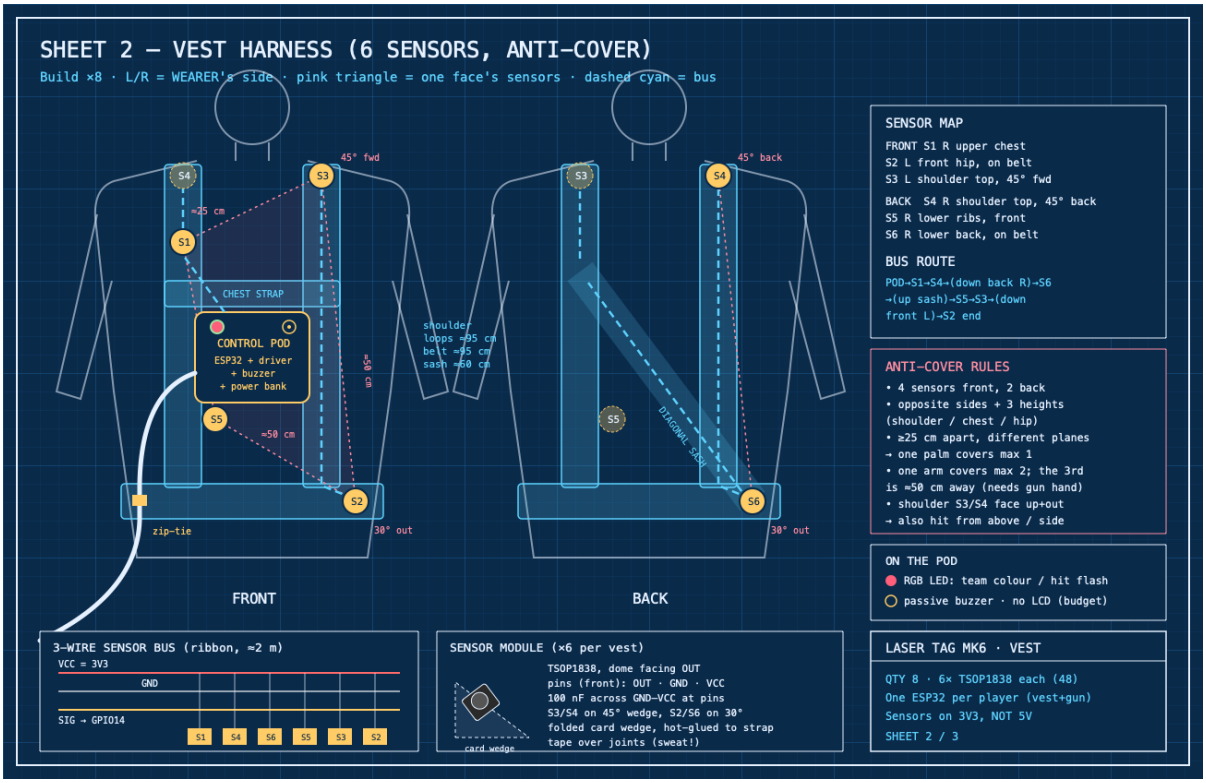
Bill of Materials (BOM)

• TSOP1838 IR Receivers (48 total) - Better range and sunlight rejection. Aim for ■15 per piece.
• 5mm 940nm IR LEDs (8 total) - Standard infrared LEDs for the guns.
• Trigger Push Buttons (8 total) - Standard 12mm tactile switches or arcade buttons.
• Passive Buzzers (8 total) - 5V passive piezos for hit sounds.
• Resistors (100Ω & 10kΩ) - 100Ω for IR LED direct drive. 10kΩ for button pull-downs.
• Ribbon Wire / Jumper Wire (1 Roll) - For wiring the 6 sensors across the body.
• Gun Housing (PVC Pipe) - 1 Length (10 ft) cut into 1-foot sections.
• Vest Harness (Nylon Webbing) - 1 Roll (20 meters) to create 'X' harnesses.

TOTAL for 8 Player Sets: ~■ 1,184 (Without transistors)

1. Vest Sensor Layout (Anti-Cover Design)

To prevent players from cheating by covering all front or back sensors at once, the 6 sensors are distributed across different heights and horizontal axes. A cross-body sash (or X-harness) is the cheapest and most effective way to mount these. 4 Front, 2 Back. Spaced out (Shoulders, Chest, Opposite Hip).



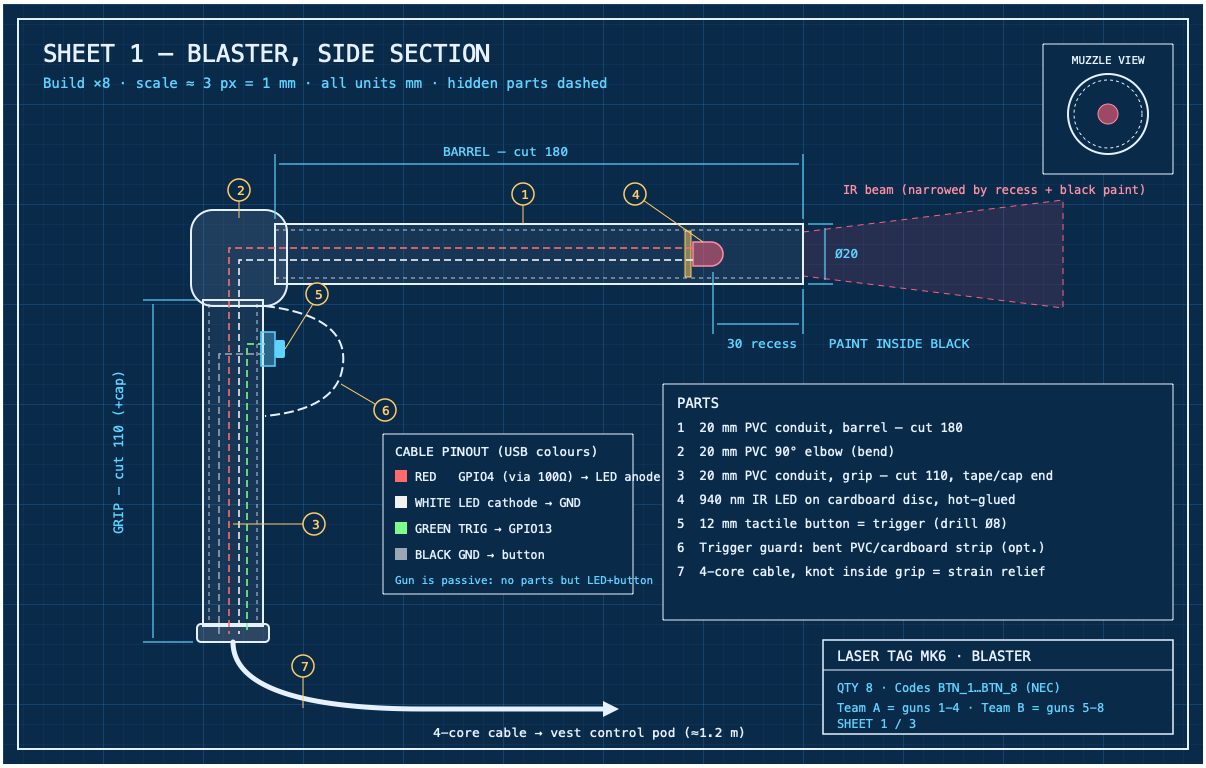
2. Gun Design (Wired to Vest)

Instead of a separate ESP32 or remote, the gun is simply a physical housing (PVC pipe) holding the IR LED and Trigger button, wired directly back to the main ESP32 on the vest.

To create a 'Real Gun' strict line-of-sight beam (No wall-bouncing):

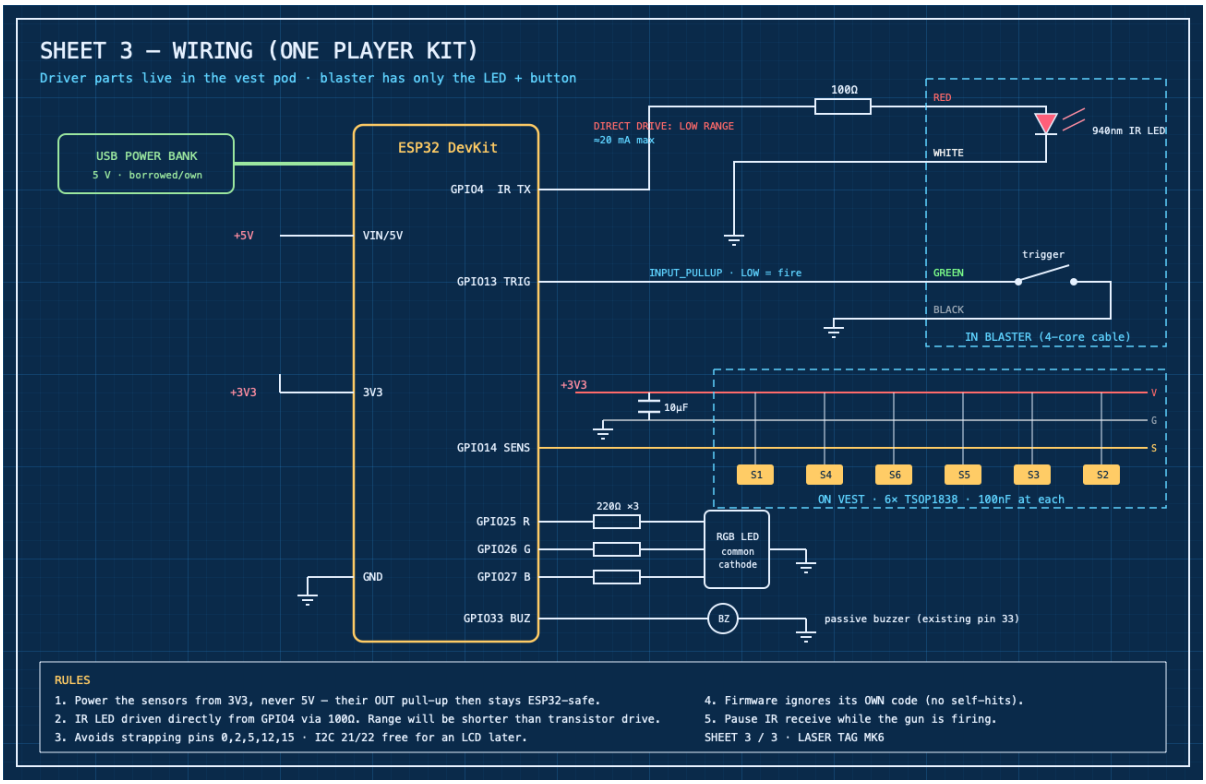
• Recess the LED: The IR LED must be glued at least 30mm deep inside the PVC pipe.

• Paint the Barrel Inside: The inside of the white PVC pipe must be painted matte black. If left white, the IR light will reflect down the tube and scatter everywhere like a shotgun.



3. Wiring Schematic

This outlines how everything connects to a single ESP32 per player. All 6 IR sensors share the same data pin.



Component Pinout Reference:

- IR Receivers (Data): Pin 15 (All 6 connected in parallel)
- IR LED (Direct via 100Ω Resistor): Pin 4
- Trigger Button: Pin 13 (with INPUT_PULLUP)
- Passive Buzzer: Pin 33